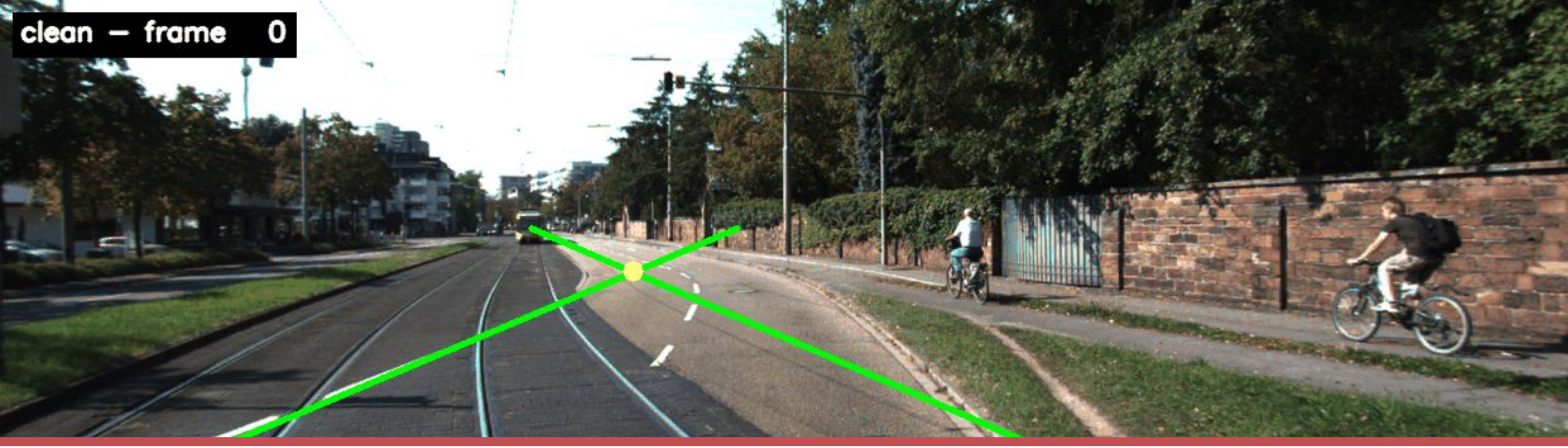


# Counterfactual Stability Analysis of Classical Lane Perception

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# Autonomous perception is evaluated frame-by-frame.

# But it operates sequentially.

Per-frame and temporal failure modes can decouple — and per-frame metrics systematically miss the latter.

*Standard lane-detection benchmarks (TuSimple, CULane) evaluate per-frame F1, recall, and accuracy* [1, 2]*.*
*Temporal consistency is rarely measured.*

# Approach

## PIPELINE

Classical CV on KITTI[3]
(Canny[4] > Hough[5] > lane fit > vanishing point)

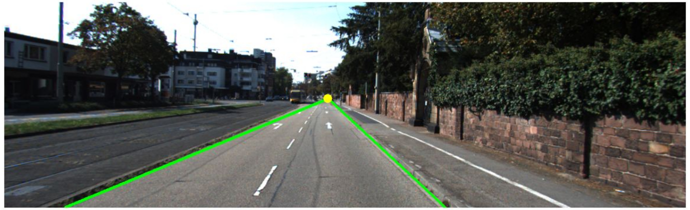
## PERTURBATIONS

Synthetic shadow, motion blur — different failure mechanisms

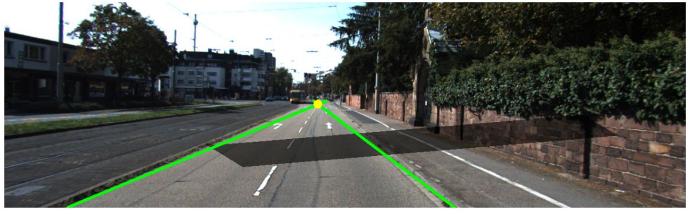
## METRICS

Five stability signals incl. VP jitter, slope oscillation, cumulative drift

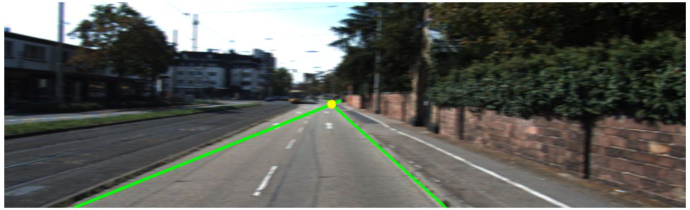
Clean baseline



With synthetic shadow ($\alpha = 0.4$)



With horizontal motion blur (k=9)

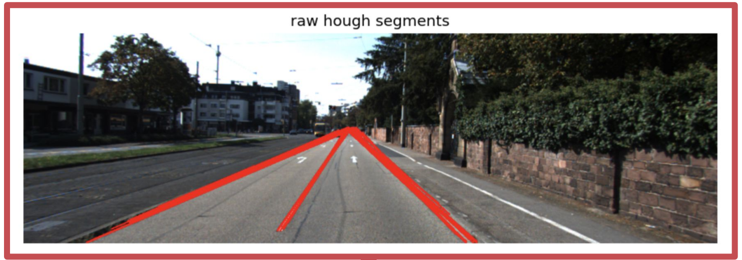


1

CANNY


2

HOUGH




3

LANE FIT
+ VP

# Baseline

## Pipeline works on clean sequences!


**100%** VP recovery

**9.55 px** *mean frame-to-frame jitter*

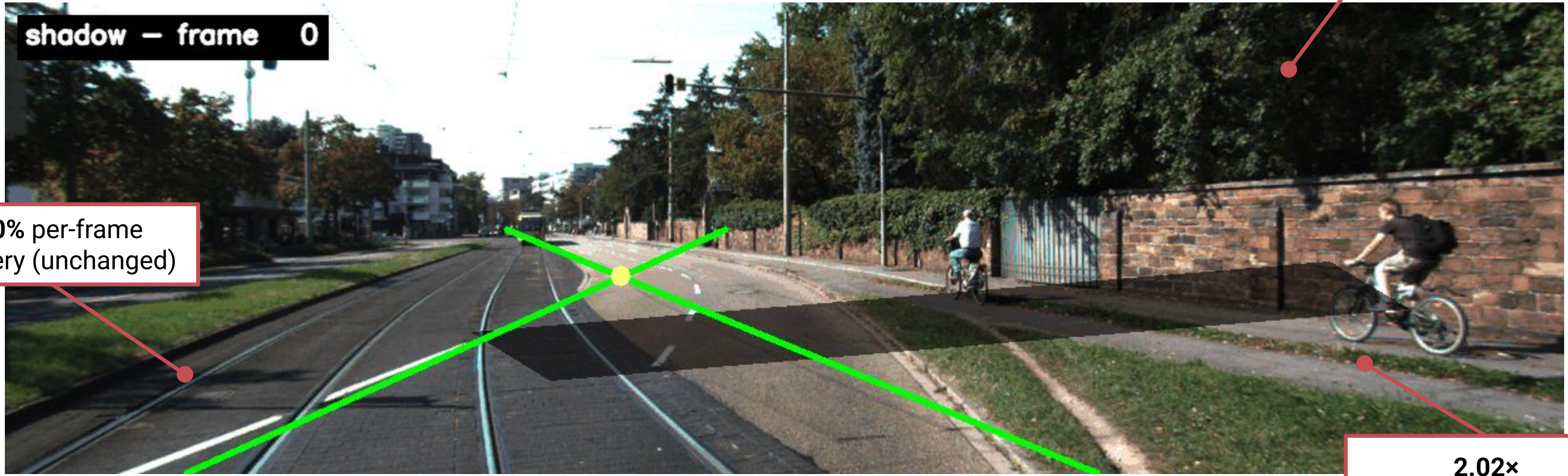
*The recovered VP traces a smooth low-frequency trajectory through the scene — our noise floor.*

# Perturbation – SHADOW

## Silent temporal failure

**200+ px** single-frame VP excursion at frame 57

**100%** per-frame recovery (unchanged)

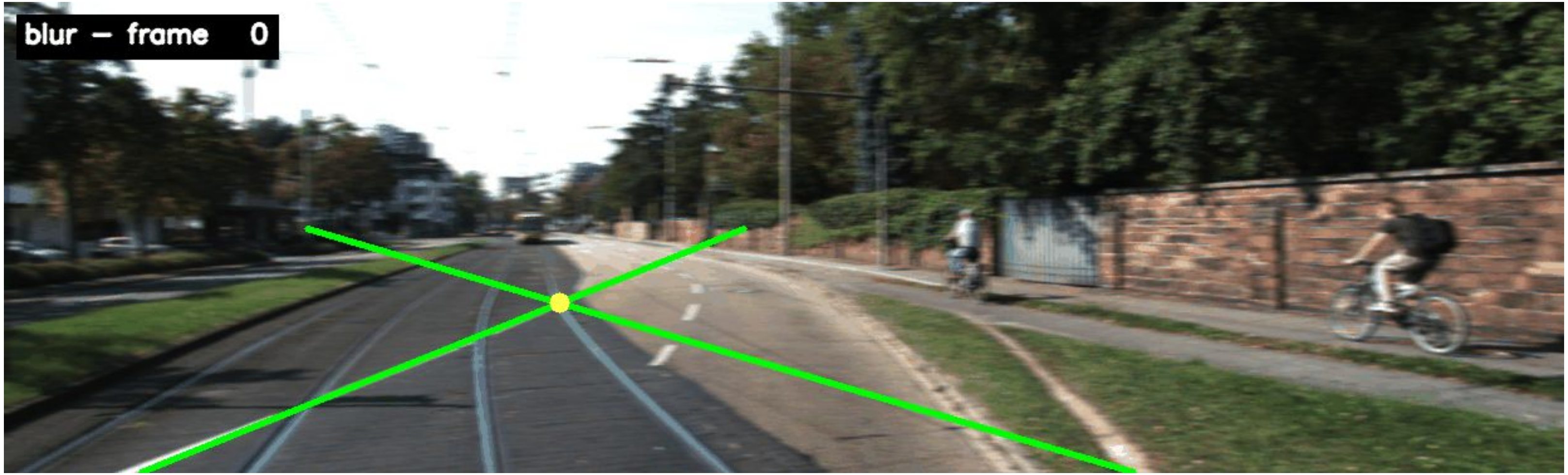



**2.02×**
*frame-to-frame jitter*

*Per-frame detection still succeeds. Invisible to most static evaluation metrics.*

# Perturbation – **MOTION BLUR**

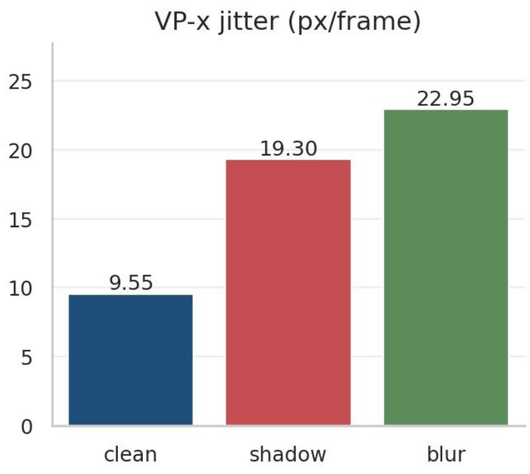
## A different failure mode

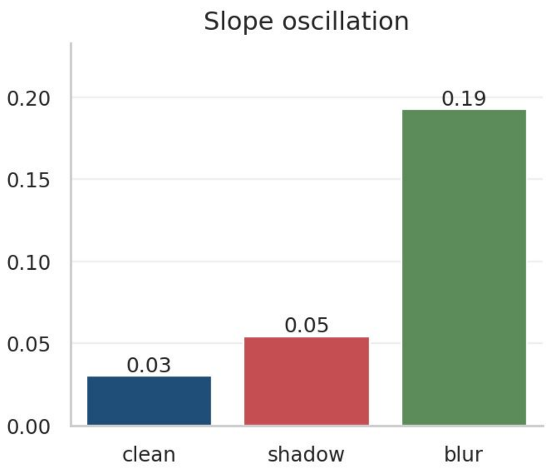



Recovery drops to **67.5%** — lane fits flicker[6]

# Perturbation – MOTION BLUR

Slope osc. **6.4× baseline** — angular geometry catastrophic, VP only mildly affected.

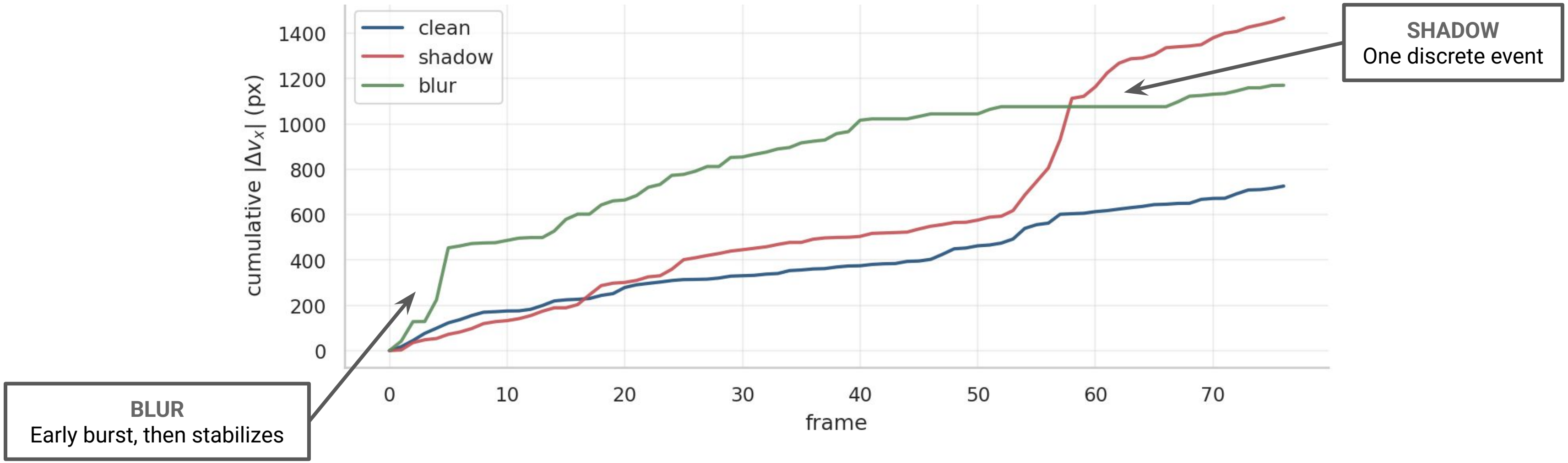



**No single metric captures both perturbations.**

# Does instability compound?

**Not monotonically.** Perturbations cause discrete events, not gradual drift.




Uncertainty signals should catch **spikes,** not smooth them.

# Takeaways

1. **Temporal inconsistency is its own uncertainty signal[7].** It carries information per-frame metrics miss.
2. **No single metric captures all perturbations.** Effective monitoring spans position, angle, and trajectory dimensions.
3. **Failures are transient events, not gradual drift.** Smoothing-based monitors will miss them.

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## NEXT STEPS

Learned pipelines · real-time monitoring · additional perturbations (lane fading, low-light, partial occlusion)

# References


[1] X. Pan et al., "Spatial as deep: Spatial CNN for traffic scene understanding," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2018.

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[3] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012.

[4] JJ. Canny, "A computational approach to edge detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 8, no. 6, pp. 679–698, 1986.

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[6] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *International Conference on Learning Representations (ICLR)*, 2019.

[7] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?" in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

## NOTE

Animated GIFs included in the presentation appear as static images in the PDF export. For access to the original animated content, please view the accompanying slide deck:

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